

22**C**

$$R_{old} = \frac{\rho L}{A} = \frac{4\rho L_{old}}{\pi d_{old}^2} = 2.000 \text{ k}\Omega$$

$$R_{new} = \frac{4\rho L_{new}}{\pi d_{new}^2} = \frac{4\rho(1.004L_{old})}{\pi(0.99d_{old})^2} = 2.049 \text{ k}\Omega$$

23**B**

Due to symmetry, **A** and **C** will have the same effective resistance, so they